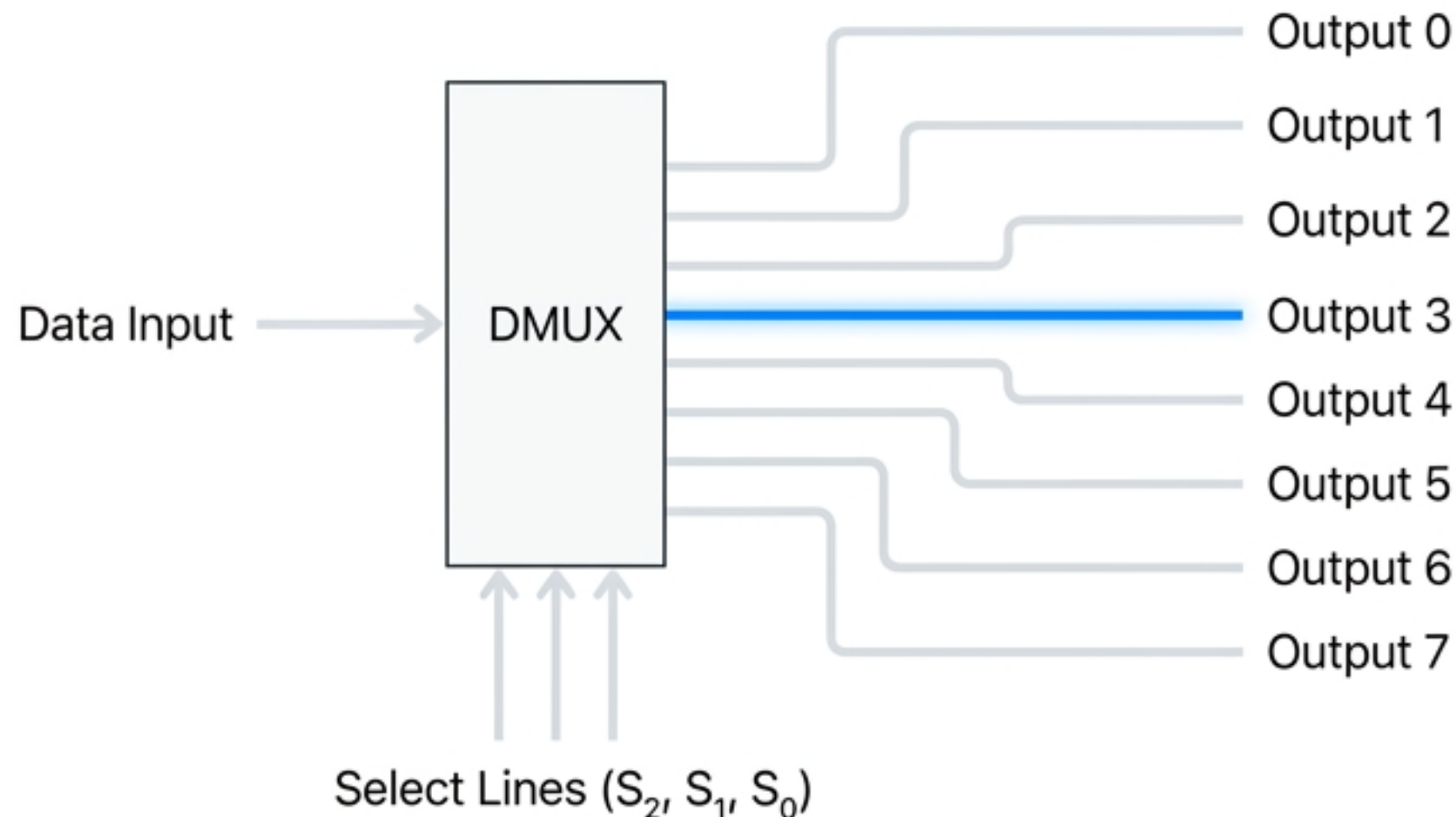


# The DEMUX: Digital Data Distributor

A Demultiplexer routes exactly 1 input signal to 1 of  $2^n$  possible output lines. The specific output path is chosen by 'n' select lines.

- Outputs =  $2^n$
- n = number of select lines.



Problem: If a DEMUX has 3 select lines ( $n = 3$ ), how many output lines will it have?

Step 1: Identify select lines ( $n = 3$ ).

Step 2: Apply formula (Outputs =  $2^n$ ).

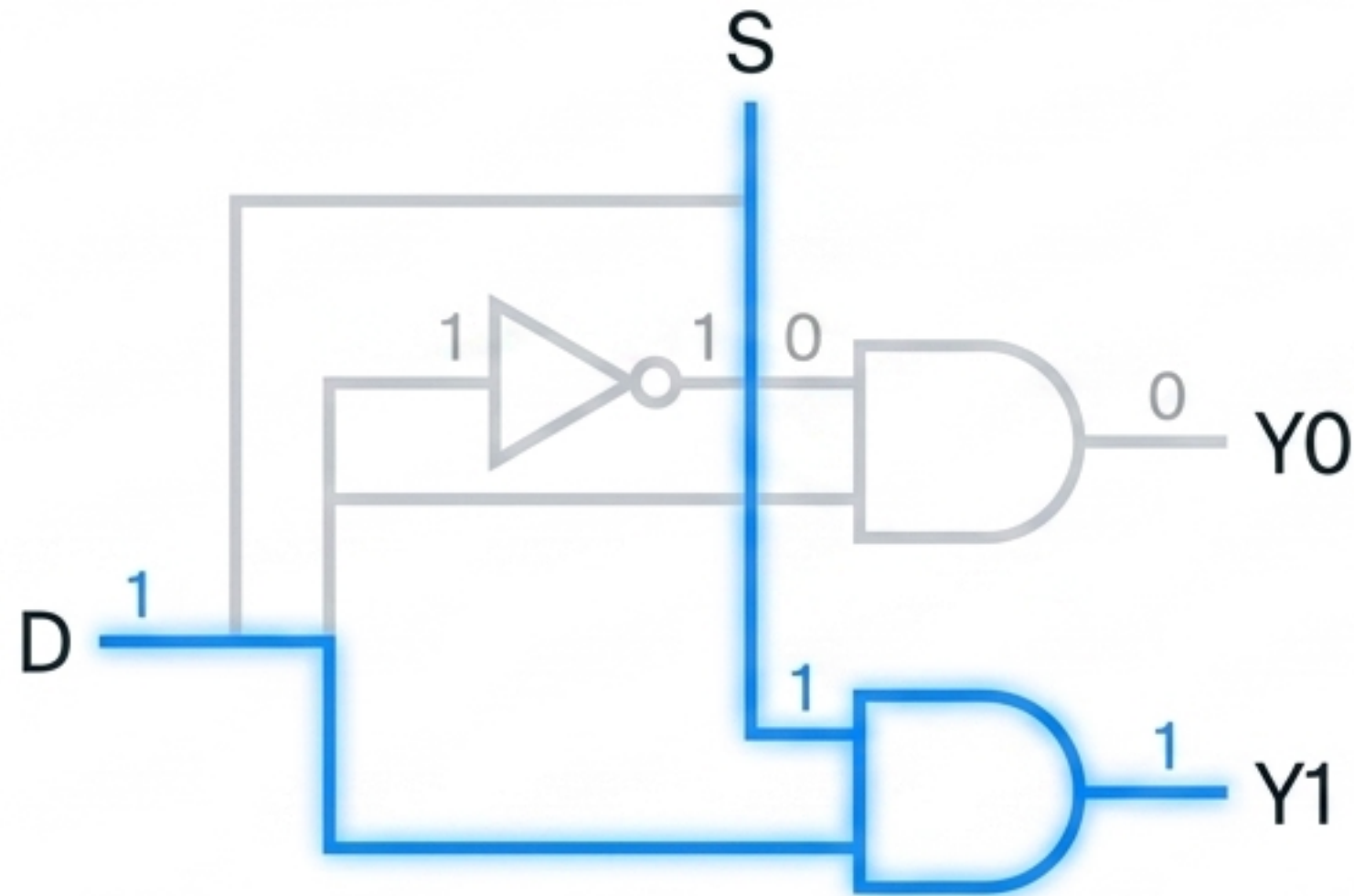
Step 3: Calculate  $2^3$  ( $2 * 2 * 2$ ).

Result: 8 outputs. This forms a 1-to-8 DEMUX.

# 1-to-2 DEMUX Architecture

$$Y0 = \text{NOT}(S) \text{ AND } D$$

$$Y1 = S \text{ AND } D$$



Problem: Given Input  $D = 1$  and Select  $S = 1$ . Find outputs  $Y0$  and  $Y1$ .

Step 1: Evaluate  $Y0$ . Substitute  $S = 1$ ,  $D = 1$  into  $Y0 = \text{NOT}(S) \text{ AND } D$ .

Step 2:  $Y0 = \text{NOT}(1) \text{ AND } 1$ . Therefore,  $Y0 = 0 \text{ AND } 1 = 0$ .

Step 3: Evaluate  $Y1$ . Substitute  $S = 1$ ,  $D = 1$  into  $Y1 = S \text{ AND } D$ .

Step 4:  $Y1 = 1 \text{ AND } 1$ . Therefore,  $Y1 = 1$ .

Result:  $Y0 = 0$ ,  $Y1 = 1$ . The data (1) is routed to  $Y1$ .



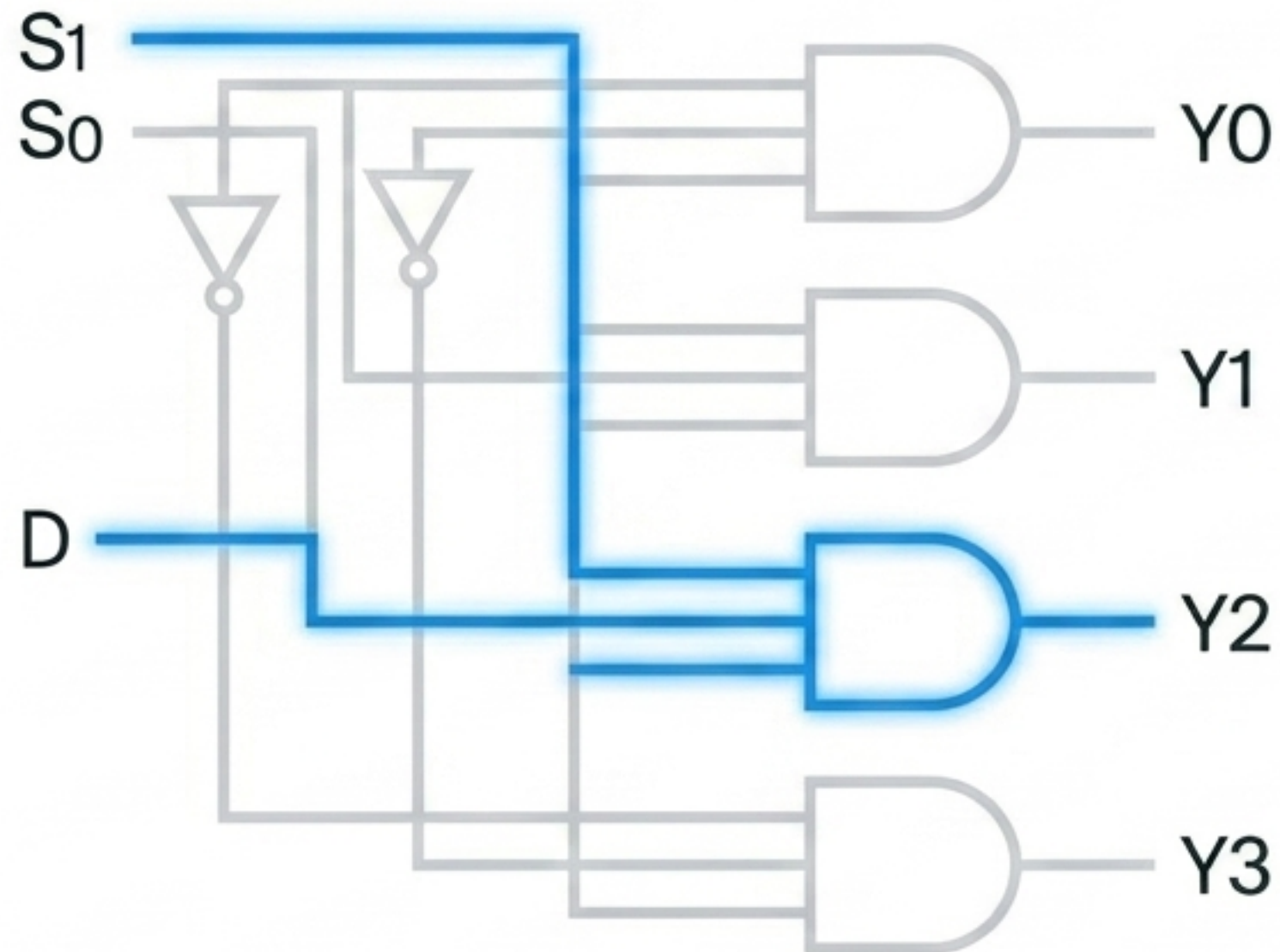
# Scaling Up: 1-to-4 DEMUX

$$Y0 = S1' S0' D$$

$$Y1 = S1' S0 D$$

$$Y2 = S1 S0' D$$

$$Y3 = S1 S0 D$$



Problem: Given Input  $D = 1$ , Select  $S1 = 1$ , Select  $S0 = 0$ . Evaluate  $Y2$ .

Step 1: Identify the equation for  $Y2$  ( $Y2 = S1 \text{ AND NOT}(S0) \text{ AND } D$ ).

Step 2: Substitute the given values ( $Y2 = 1 \text{ AND NOT}(0) \text{ AND } 1$ ).

Step 3: Resolve the NOT operator ( $\text{NOT}(0) = 1$ ).

Step 4: Evaluate the AND operations ( $Y2 = 1 \text{ AND } 1 \text{ AND } 1$ ).

Result:  $Y2 = 1$ . Data successfully routed.

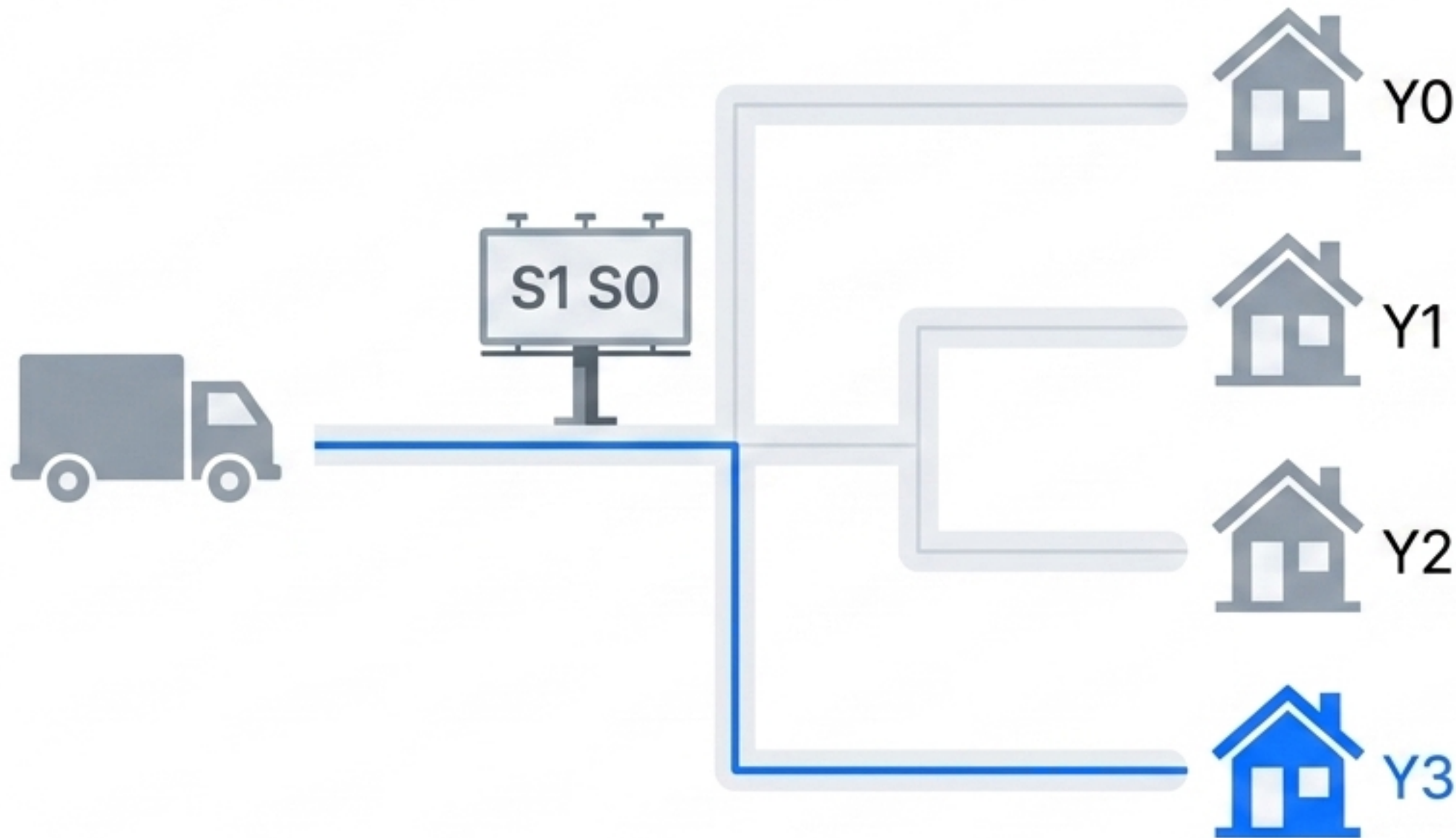


# The Parcel Delivery Analogy

**Input (D)** = 1 Parcel

**Outputs (Y0 to Y3)** = 4 Houses

**Select Lines (S1, S0)** = 2 Digit Binary Address



**Problem:** A Parcel ( $D = 1$ ) arrives. The routing Address is  $S1 = 1$ ,  $S0 = 1$ . Find the destination.

**Step 1:** Read the binary Address (1 1).

**Step 2:** Convert binary (1 1) to decimal ( $1 * 2^1 + 1 * 2^0$ ).

**Step 3:** Calculation =  $2 + 1 = 3$ .

**Result:** The Parcel ( $D = 1$ ) is delivered to House Y3.